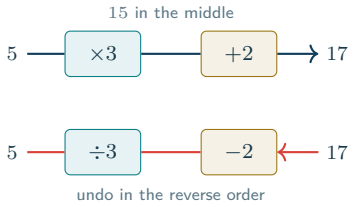
ALGEBRA LADDER 3 OF 9

Identity, inverse and undoing

Every operation has a do-nothing element and a way to reverse it, and solving an equation is the art of undoing.

1 / THE RULE AND THE PICTURE

An identity changes nothing, and an inverse brings you back to the identity



Adding 0 and multiplying by 1 leave a number alone, and every number can be brought back to those do-nothing values by an inverse:

$$a + 0 = a, \quad a + (-a) = 0, \quad a \times 1 = a, \quad a \times \tfrac{1}{a} = 1$$

The machine above takes 5 to 17 by tripling and adding 2, and the undo machine reverses the steps in the opposite order.

2 / IDENTITIES AND INVERSES

Each operation has its own pair

operation	identity	inverse of a	what the inverse does
addition	0, since $a + 0 = a$	$-a$	subtracting a
multiplication	1, since $a \times 1 = a$	$1/a$, for $a \neq 0$	dividing by a
doing one step then another	doing nothing	the step that reverses it	the undo machine

3 / SOLVING AN EQUATION IS UNDOING

Undo the last step first and keep the order reversed

To solve $3x + 2 = 17$ the machine triples and then adds 2, so we subtract 2 to get 15 and divide by 3 to get $x = 5$, which checks since $3 \times 5 + 2 = 17$.
For $\frac{x}{4} - 1 = 6$ the machine divides by 4 and subtracts 1, so we add 1 to get 7 and multiply by 4 to get $x = 28$, and indeed $\frac{28}{4} - 1 = 6$.
Doing the same thing to both sides of an equation is the balance method, and it is the undo machine written line by line.

4 / SUBTRACTION AND DIVISION IN DISGUISE

Taking away is adding the inverse, and dividing is multiplying by the reciprocal

Writing $8 - 3 = 8 + (-3)$ and $12 \div 4 = 12 \times \frac{1}{4}$ explains why the two laws failed on the first sheet: the inverse hides in front of the second number, so $3 - 8 = 3 + (-8) = -5$ and the order matters. In $10 - (4 - 3)$ the bracket flips the sign of the 3, giving $10 + (-4) + 3 = 9$, while $(10 - 4) - 3 = 10 + (-4) + (-3) = 3$.
With the inverse made visible only the commutative and associative laws of addition are in use.

5 / WHY THE NUMBERS GROW

Inverses that are missing force new kinds of number

Among the counting numbers $3 - 5$ has no answer, so the negative numbers are invented to give every number an additive inverse, and among the integers $3 \div 4$ has no answer, so fractions are invented to give every number except zero a reciprocal.
Zero has no reciprocal because $0 \times x = 0$ for every x, so no number multiplies with 0 to give 1, which is why $\frac{1}{0}$ is left undefined.

6 / TRY THESE

Find the inverse and undo the machine

(a) The inverse of $\div 5$ is $\times 5$, and the inverse of "add 7" is "subtract 7".
(b) Solve $4x - 3 = 21$: add 3 to get 24, divide by 4 to get $x = 6$.
(c) A machine squares and then adds 1: why can it not be undone uniquely? Both 3 and -3 give 10, so the undo machine would need to choose, which is why square roots carry a $\pm$.

ASIDE / THE SHAPE OF THE LADDER

A set with one operation, an identity and inverses is called a group, and the numbers under addition are the first example.

LINKS TO OTHER SHEETS

The previous sheet uses subtraction and division as examples of failing laws; the composition sheet returns to inverses that reverse the order of steps.

Convention. The inverse of a under addition is $-a$ and under multiplication is $1/a$; an equation is solved by applying inverses in reverse order.